Explicit Depth-Aware Blurry Video Frame Interpolation Guided by Differential Curves

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Abstract

Blurry video frame interpolation (BVFI), which aims to generate high-frame-rate clear videos from low-frame-rate blurry inputs, is a challenging yet significant task in computer vision. Current state-of-the-art approaches typically rely on linear or quadratic models to estimate intermediate motion. However, these methods often overlook depth variations that occur during fast object motion, leading to changes in object size and hindering interpolation performance.This paper proposes the Differential Curves-guided Blurry Video Frame Interpolation (DC-BVFI) framework, which leverages the differential curves theory to analyze and mitigate the effects of depth variations caused by object motion. Specifically, DC-BVFI consists of UBNet and MPNet. Unlike prior approaches that rely on optical flow for frame interpolation, MPNet is designed to estimate the 3D scene flow, which facilitates a more precise awareness of depth and velocity variations. Since scene flow cannot be directly inferred in the 2D frame space, UBNet is introduced to transform them into 3D point maps. Extensive experiments demonstrate that the proposed DC-BVFI framework surpasses state-of-the-art performance in simulated and real-world datasets.

1. Introduction

Natural videos captured by consumer cameras often suffer from real-world movement complexity. Blurry Video Frame Interpolation (BVFI) enhances low frame rate and motion-blurred videos into high-quality, sharp outputs between consecutive input frames by reconstructing motion,

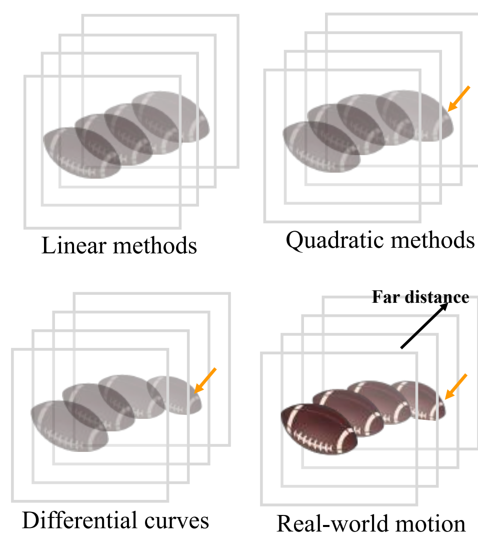

Figure 1. Challenge. Linear methods: Synthesizing intermediate frames through linear approximation. Quadratic methods: Synthesizing intermediate frames by fitting velocity variations. Differential curves: Exploring the depth variations.

thereby improving perception for the human visual system [22, 23]. Therefore, it is widely used for diverse applications[20, 34, 49], and it has become a significant research area in computer vision.

BVFI methods can be broadly divided into two categories: linear methods and quadratic methods. Linear methods [3, 12, 29, 30, 32] typically employ existing motion estimators [17, 43, 45] to predict optical flow between input frames. The intermediate optical flow is obtained by linearly approximating the optical flow between input frames to synthesize the intermediate frame. However, linear methods often overlook variations in the velocity of object motion. To more accurately reconstruct complex motion, un-

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like linear methods, quadratic methods [11, 28, 50, 51, 56] leverage optical flow to explore velocity variations in videos for fitting complex motions.

Unfortunately, upon scrutinizing and experimenting on the released implementations of quadratic methods[11, 27, 28, 51], we observed that they overlook the depth variations and size changes caused by the car's motion, making them ineffective for video frame interpolation. We conjecture that the reasons are twofolds. **Limitations of the Quadratic Theory.** The quadratic theory primarily focuses on velocity variations in the horizontal and vertical directions, overlooking the object size variations caused by depth variations in the real world. Specifically, the quadratic theory is based on two linearly independent vectors and lacks a linearly independent vector to represent depth variations. As a result, when an object's size changes, the quadratic theory fails to describe it. **Limitations of the Optical Flow.** The implementation of the quadratic theory relies on optical flow; however, optical flow struggles to capture motion-induced depth variations. Specifically, an intrinsic limitation of optical flow is that a nearby object that moves slowly will generate the same optical flow as a faraway object that moves fast [5, 52].

Based on the above analysis, to overcome the limitations of the quadratic theory in handling depth variations, we propose leveraging the differential curves theory to unify the modeling of depth and velocity variations in motion. Specifically, we map pixels onto slight curve segments in 3D space, with each curve defined by a unit basis and motion parameters. The unit basis represents spatial coordinates in 3D space, while the motion parameters describe the 3D displacement of the unit basis for the next time step. Differential curves provide an intuitive core idea for describing 3D spatial motion; however, its implementation presents certain challenges. The primary challenges include converting 2D video frames into 3D point map representations and accurately estimating the motion parameters between point maps.

To convert 2D video frames into 3D point maps, we propose **UBNet**. Although the idea of converting video frames into 3D camera space point maps is intuitive and sensible, the task is challenging because it requires estimating camera parameters and monocular depth maps from video frames. To address this issue, UBNet leverages a general transformer architecture [46], which [37, 46, 47, 59] can estimate camera intrinsics, depth maps, and camera-space point maps from unposed frames.

To estimate motion parameters between point maps, we introduce **MPNet**. MPNet estimates motion parameters with depth awareness, but this task is challenging because it requires estimating scene flow from the desired point map to the source point map, which existing methods struggle to achieve. To address this issue, MPNet combines a method that upgrades the 2D optical flow obtained from the video interpolation model (IFnet [14]) to produce the camera-space scene flow. By leveraging these modules, we build a Differential Curves-guided Blurry Video Frame Interpolation (**DC-BVFI**) framework.

This paper presents three key contributions: **(1)** To the best of our knowledge, this is the first work that introduces differential curves theory to alleviate the challenge of depth variations in blurry video multi-frame interpolation. **(2)** This paper proposes an explicit depth-aware video interpolation framework to overcome depth variation and synthesize intermediate video frames more effectively. The UBNet converts video frames into explicit 3D point maps in camera space, and the MPNet estimates the scene flow between these point maps. **(3)** DC-VFI outperforms state-of-the-art results across four benchmark datasets.

2. Related Work

Video multi-frame interpolation. Recently, advancements in deep learning have led to various methods for video frame interpolation. These methods can be broadly categorized into two main paradigms: reconstruction-based and generation-based.

Reconstruction-based methods are mainly categorized into two types: optical flow-based methods and kernel-based methods. Initially, optical flow-based methods introduced a linear prior[3, 12, 29, 30, 32], which were subsequently expanded to quadratic flow prior [11, 28, 51, 55] and even cubic [6] trajectory models to represent the local motion of pixels accurately. To obtain the optical flow from the intermediate frame to the input frames, [11, 13] proposed distillation strategies to obtain the optical flow from the intermediate frame to the input frames. These Kernel-based methods are implemented as separable convolutions [35], deformable convolutions [4, 25, 26, 40], and trilinear sampling kernels[30].

Methods based on Denoising Diffusion Probabilistic Models (DDPM)[10] leverage generative techniques like DDPM to fill occlusions caused by motion. These DDPM-based approaches are implemented as score-based diffusion [8, 19, 48], motion-aware diffusion [15], and Brownian bridge diffusion [24, 31]. However, most DDPM-based methods are significantly time-consuming and challenging for real-time inference.

Blurry video multi-frame interpolation. When the original reference frames degrade, existing VFI methods face significant challenges in recovering sharp intermediate frames and reconstructing motion trajectories. Early methods primarily employed a cascade of deblurring and interpolation techniques [9, 21]. [21, 38, 39] extract frames, which are then subsequently used to generate intermediate clear frames by jointly optimizing. Oh et al. [36] proposed a DeMFI for joint deblurring and multi-frame interpolation, incorporating a flow-guided attentive-correlation-based feature bolstering module and recursive boosting. The critical contribution of Zhong et al. [58] is contributing a real-world blurry video with a low frame rate dataset.

3. Method

3.1. Preliminary of Differential Curves

According to the definition of differential curves [44], the unit basis can be written as:

$$\begin{cases} \alpha(s) = r^{'}(s), \\ \beta(s) = r^{''}(s), \\ \gamma(s) = \alpha(s) \wedge \beta(s), \end{cases} \tag{1}$$

where $'$ is the derivative symbol of $r(s)$, $''$ is the second derivative symbol of $r(s)$, and $\wedge$ is the symbol of the outer product. The $r(s)$ is a slight segment curve of the point s. The $\alpha(s)$, $\beta(s)$, and $\gamma(s)$ together represent the spatial position of $r(s)$, termed the **unit basis**.

According to the definition of differential curves [44], the derivative of the unit basis can be written as:

$$\begin{bmatrix} \alpha^{'}(s) \\ \beta^{'}(s) \\ \gamma^{'}(s) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix}}_{\text{motion parameters}} \begin{bmatrix} \alpha(s) \\ \beta(s) \\ \gamma(s) \end{bmatrix}. \tag{2}$$

$\kappa(s)$ and $\tau(s)$ is the curvature and torsion. The $\kappa(s)$ and $\tau(s)$ together represent the spatial displacement of $r(s)$, termed the **motion parameters**. Essentially, the motion parameters depict the spatial displacement of $r(s)$, similar to optical flow. We simplify the notation $(\alpha(s), \beta(s), \gamma(s), \kappa(s)$ and $\tau(s))$ to $(\alpha, \beta, \gamma, \kappa$ and $\tau)$ for convenience expression.

3.2. Problem formulation

Blurry Video Frame Interpolation. In the blurry video frame interpolation task, it can be written as

$$I_t = \mathbf{F}(B_i, B_{i+\eta}), t \in (i, i+1, \ldots, i+\eta), \tag{3}$$

where B_i and $B_{i+\eta}$ are input frames. i is a current time index and t is an interpolated time index between i and $i+\eta$. The goal is to synthesize a frame I_t by inputting frames $B_i, B_{i+\eta}$ and an optimization function $(\mathbf{F}(\cdot))$.

Blurry Video Frame Interpolation under Differential Curves. Most failures in video frame interpolation occur due to object motion, which inevitably leads to depth changes (object scale changes) between adjacent frames. To mitigate the impact of depth variation on video interpolation quality, we redefine the video interpolation problem under differential curves theory, allowing for explicit exploration of depth changes.

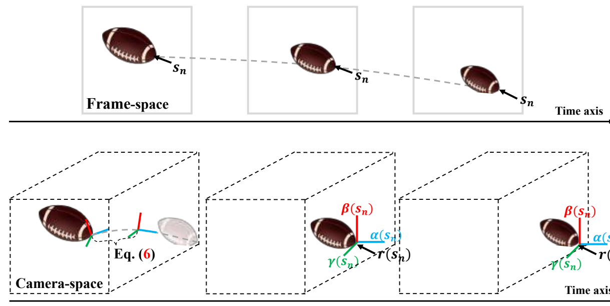

Figure 2. Video Frame under differential curves. From the perspective of differential curves, each pixel s_n can be viewed as a light segment curve $r(s_n)$.

For the convenience of expression, we write a time sequence $B_i \rightarrow I_t$. Each frame B_i contains $H \times W$ pixels $(s_n, n = 1, \ldots, H \times W)$. From the perspective of differential curves, each pixel s_n can be viewed as a slight segment curve $r(s_n)$, where its spatial position denotes as $[\alpha(s_n), \beta(s_n), \gamma(s_n)] \in \mathbb{R}^{1\times3}$. Then, the frame B_i can be presented as $H \times W$ slight segment curves, denoted as $[\alpha, \beta, \gamma]_i \in \mathbb{R}^{H\times W\times3}$. Subsequently, the video frame interpolation process $(\mathbf{F}(B_i) \rightarrow I_t)$ can be reformulated as $(\mathbf{F}([\alpha, \beta, \gamma]_i) \rightarrow [\alpha, \beta, \gamma]_t)$. We reformulate the blurry video frame interpolation problem under the differential curves theory and write as follows:

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_t = \mathbf{F}\left(\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_i\right), t \in (i, i+1, \ldots, i+\eta) \tag{4}$$

Our goal is $F([\alpha, \beta, \gamma]_i) \rightarrow [\alpha, \beta, \gamma]_t$. $\mathbf{F}(\cdot)$ is an optimization function that predicts the position at time t from the position at time i through optimization algorithm.

3.3. Optimization algorithm

A multi-stage strategy ensures the temporal continuity of motion [27]; however, current multi-stage strategies[27, 36, 38, 39] lack algorithmic convergence, making it challenging for them to converge quickly. To swiftly and directly approach the target, we propose an optimization algorithm. Following Eq. (4), the minimized objective function can be written as

$$\min \left\| \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_t - \mathbf{F}\left(\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_i\right) \right\|_2^2. \tag{5}$$

To solve the minimization Eq. (5), we utilize the gradient descent algorithm [1].

Leveraging the gradient descent algorithm [1], the Eq. (5) can be optimized:

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_i^2 = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_i^1 + \underbrace{\begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix}_i^1 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}_i^1}_{\text{gradient } (SF_{i\rightarrow t})}. \tag{6}$$

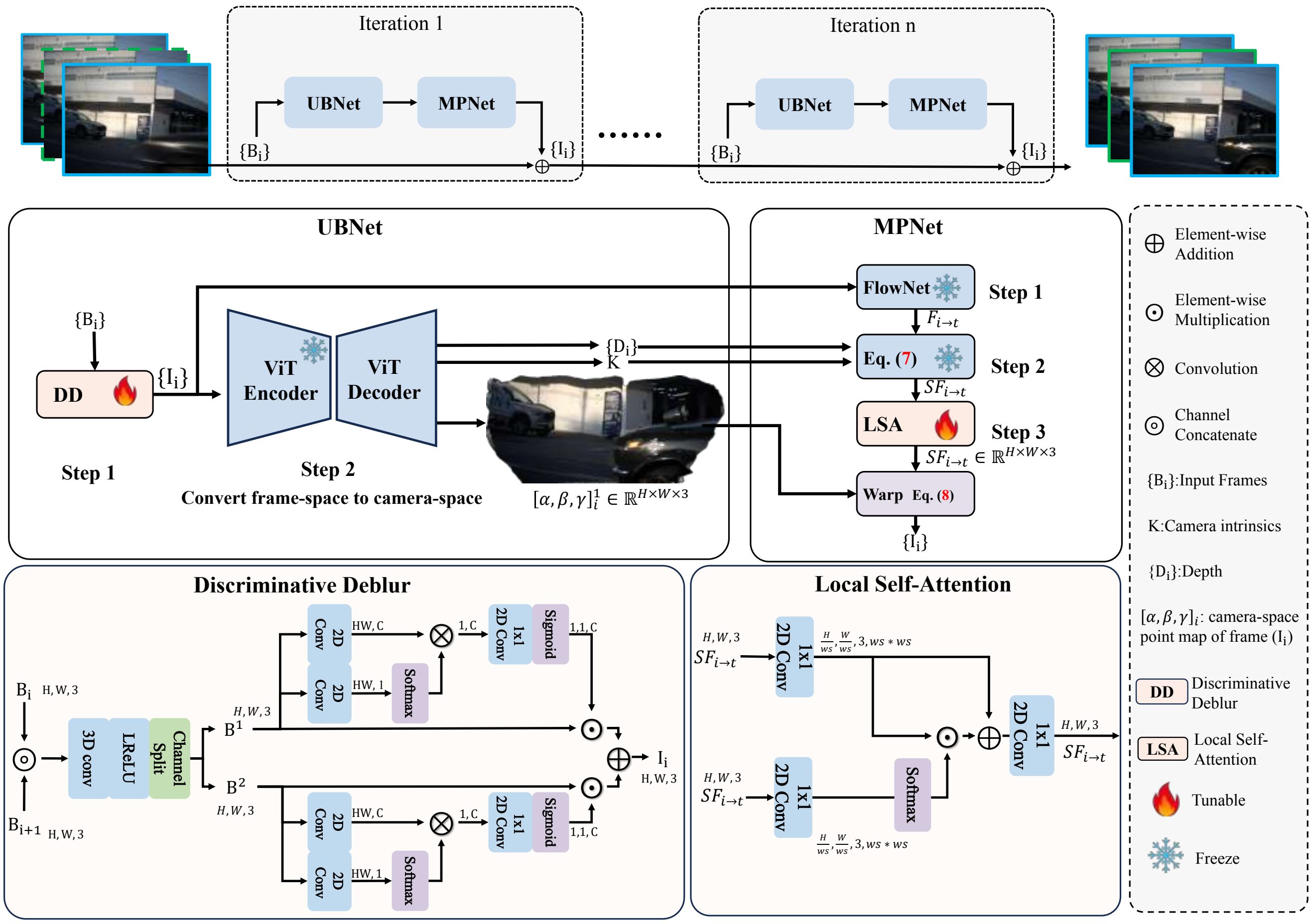

Figure 3. Overview of the proposed DC-BVFI. UBNet primarily converts video frame I_i into camera space point map $[\alpha,\beta,\gamma]_i \in \mathbb{R}^{H\times W\times 3}$, where 3 represents the 3D coordinate, and $H \times W$ indicates the total number of point. MPNet estimates depth-aware scene flow $SF_{i\to t} \in \mathbb{R}^{H\times W\times 3}$, where 3 represents the 3D coordinate offset between two point maps.

$[\alpha,\beta,\gamma]_i^1$ denotes the spatial position of $H \times W$ slight segment curves at the 1^{st} iteration. And $[\alpha,\beta,\gamma]_i^2$ is the spatial position of the same slight segment curves at the 2^{nd} iteration. $[\alpha,\beta,\gamma]_i^1$ is updated by adding the gradient of $[\alpha,\beta,\gamma]_i^1$. Figure 2 shows the iterative optimization process from 1^{st} iteration to 2^{nd} iteration (Eq. (6)).

3.4. Differential Curves for Blurry Video Frame Interpolation

Although using differential curves theory to describe spatial motion is reasonable, its implementation is challenging: 1) How to convert a video frame I_i into a camera space point map $[\alpha,\beta,\gamma]_i$ and represent the spatial position. 2) How to estimate the relative displacement SF from an unknown point map to a source point map. And existing methods struggle to obtain the relative displacement. To this end, UBNet and MPNet are designed in DC-BVFI to address these issues. The pipeline of the proposed DC-BVFI is outlined in Fig. 3.

3.4.1. UBNet

Movement inevitably causes motion blur [38] and depth change. To convert the blurred video frame B_i into a 3D camera space point map $[\alpha,\beta,\gamma]_i$, we designed a UBNet. More specifically, the UBNet includes two submodules:

Setp 1. The function of this submodule is to deblur. Traditional methods often simply apply Direct Stacking of video frames for deblurring. However, motion usually causes the content of frames to be different, resulting in misalignment between frames. Directly Stacking (DS) misalignment frames can hinder intermediate frame restoration, which may not facilitate video deblurring. Therefore, we introduce a Discriminative Deblur (DD) strategy, as shown in Fig. 3, to mitigate the adverse effects of blurring. The operations can be expressed as follows: $\{B_i\} \xrightarrow{\text{DD}} \{I_i\}$.

Setp 2. Real-world motion is a three-dimensional movement involving both depth and velocity variations. However, existing video frame interpolation methods typically focus only on velocity variations, which limits the quality of interpolation. To effectively explore three-dimensional motion, it is essential to convert the two-dimensional pixel space into the three-dimensional camera space. To convert the two-dimensional pixel space into the three-dimensional camera space, we employ a general 3D reconstruction paradigm [46], which can estimate camera intrinsics, depth maps, and point maps in the camera space from video frames. Specifically, the coordinate of the frame is mapped to a three-dimensional coordinate $[\alpha, \beta, \gamma] \in \mathbb{R}^{H\times W\times 3}$, where 3 represents the 3D coordinate, and $H \times W$ indicates the total number of point. The operations can be expressed as follows: $\{I_i\} \xrightarrow{\text{ViT}} \{[\alpha, \beta, \gamma]_i\}$.

3.4.2. MPNet

In optical flow-based video interpolation methods, it is often necessary to estimate the optical flow $F_{i\to t} \in \mathbb{R}^{H\times W\times 2}$ from the frame I_i to the frame I_t for interpolation [13, 28, 51]. Similarly, in 3D camera space, it is also necessary to estimate the scene flow from the point map $[\alpha, \beta, \gamma]_i$ to the point map $[\alpha, \beta, \gamma]_t$. However, because the $[\alpha, \beta, \gamma]_t$ is unknown, existing methods cannot directly obtain the scene flow. To this end, we propose MPNet to upgrade the optical flow $F_{i\to t}$ to the scene flow $SF_{i\to t}$.

Step 1. Given the frames $\{I_i\}$, flow-based methods [13, 51, 54] require approximating the intermediate flows $F_{i\to t}$ from the perspective of the synthesized frame I_t. Considering real-time performance, we employ IFNet [13] to estimate the optical flow $F_{i\to t}$.

Step 2. To express positional changes between point maps, we upgrade the optical flow $F_{i\to t}$ to camera space scene flow $SF_{i\to t}$ to aware depth variations. The process is written as the following formula:

$$SF_{i\to t} = (F_{i\to t}|D_{i\to t}) \times K^{-1}, \tag{7}$$

where $D_{i\to t} \in \mathbb{R}^{H\times W\times 1} = D_i - D_t$ is the depth maps and the camera intrinsics $K \in \mathbb{R}^{3\times 3}$. We set $D_t = (1-t)D_i + tD_{i+\eta}$, where D_i and $D_{i+\eta}$ are the depth maps of two adjacent frames. $t \in \{i, i+\eta\}$ is the time index for interpolation. The $\times$ denotes matrix multiplication, while $|$ represents concatenation. The scene flow $SF_{i\to t} \in \mathbb{R}^{H\times W\times 3}$, where 3 represents the 3D coordinate offset between two point maps. We obtain the coarse scene flow $SF_{i\to t}$.

Step 3. Blurry frames lead to rough optical flow, leading to coarse scene flow. Therefore, it is necessary to refine the coarse scene flow. To achieve this, we introduce the Local Self-Attention (LSA) and Residual Block (RB) strategies, which perform fine-grained adjustments at the pixel level to obtain refining scene flow.

3.4.3. Warp

A brief explanation of the point map warping process by scene flow. Specifically, given a scene flow $SF_{i\to t}$ and a point map $[\alpha, \beta, \gamma]_i$. The warp process can be expressed as:

$$[\alpha, \beta, \gamma]_t = [\alpha, \beta, \gamma]_i + SF_{i\to t}, \tag{8}$$

where $+$ denotes element-wise addition and $[\alpha, \beta, \gamma]_t$ represents the warped point map.

To Frame. It is important to note that the warped point map represents 3D coordinates. We need to project the point map back into the 2D coordinate system. The pseudocode for warping and projecting onto the 2D frame is as follows:

Algorithm 1 Pseudocode.

```
# Warp
pt_t = pt + sf

# point map is projected to frame
b, _, h, w = pt.size()
pt_t_ = torch.matmul(K, pt_t.view(b, 3, -1))
coord_t = pt_t_.div(pt_t_[:, 2:3, :])[:, 0:2, :]
coord_t = coord_t.view(b, 2, h, w)
grid = coord_t.transpose(1, 2).transpose(2, 3)
I_t = torch.grid_sample(I_i, grid)
```

3.5. Loss Function.

We use the mean square error (MSE) as the training loss function:

$$\mathcal{L} = \|I_i - I_i^{gt}\|^2 + \|I_t - I_t^{gt}\|^2 \tag{9}$$

The variable I_t^{gt} denotes the ground truth frame. I_t denotes the output frame of the model.

4. Experiments

4.1. Dataset

Adobe. The Adobe dataset [42] consists of 120 videos at 240 fps with a resolution of 1280 × 720. The blurry frames are 30 fps and downsized to 640 × 352 as done in [36, 38]. We follow [36, 39] to use the same 8 videos of the Adobe dataset for evaluation and other videos of the Adobe dataset for training. **RBI.** The real-world blurry frame interpolation [58] dataset comprises 55 videos. We use the same videos as the test set and the others as the training set [58]. The blurred image has an exposure time of 18 ms, while the sharp image has an exposure time of nearly 2 ms [58]. This means that there are 9 sharp frames corresponding to one blurred frame, and 11 sharp frames exist in the readout dead-time between adjacent blurred frames [58]. **GoPro.** We use the same 11 videos of the GoPro dataset [33] to test the model [38]. The test datasets are also temporally downsampled to 30 fps with the blurring as [36]. **YouTube.** As a test dataset, the YouTube [38] consists of 59 video clips with different scenes and 12590 frames in total [38]. Please note that the evaluation datasets are also temporally downsampled to 30 fps with the blurring as [36].

Table 1. Quantitative comparisons on Adobe and RBI. The best and the second best results are denoted by pink and yellow.

	Adobe						RBI					
	Deblurring		MFI		Avg		Deblurring		MFI		Avg	
Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
QVI [51]	28.73	0.865	28.66	0.864	28.67	0.864	27.04	0.840	26.92	0.838	27.13	0.840
TNTT [21]	29.40	0.893	29.45	0.876	29.45	0.876	28.88	0.857	27.36	0.854	28.11	0.855
DAIN [2]	–	–	29.30	0.877	–	–	–	–	28.38	0.860	–	–
BIN [38]	31.76	0.896	28.36	0.847	29.95	0.878	29.01	0.860	27.58	0.861	28.30	0.861
ALANET [9]	33.71	0.932	32.98	0.936	33.34	0.935	28.81	0.868	28.88	0.868	28.84	0.868
UTI-VFI [56]	28.73	0.865	28.66	0.864	28.67	0.864	27.34	0.844	27.12	0.847	27.23	0.844
EQVI [28]	–	–	29.69	0.889	–	–	–	–	27.79	0.840	–	–
ANIB [57]	31.30	0.890	32.83	0.899	32.52	0.894	28.83	0.888	28.99	0.872	28.90	0.870
DeMFI [36]	33.83	0.937	33.79	0.941	33.79	0.940	29.69	0.899	29.00	0.890	29.35	0.893
BiT [58]	33.60	0.936	33.86	0.938	33.73	0.937	29.58	0.893	29.84	0.897	29.75	0.894
JNMR [27]	31.33	0.907	31.68	0.935	31.43	0.939	28.52	0.873	28.71	0.879	28.63	0.875
IQ-VFI [11]	33.83	0.940	33.80	0.941	33.79	0.940	29.95	0.894	29.92	0.897	29.91	0.894
Ours	34.17	0.942	34.10	0.942	34.14	0.942	31.02	0.904	30.89	0.915	31.03	0.913

Table 2. Quantitative comparisons on GoPro and YouTube. The best and the second best results are denoted by pink and yellow.

	YouTube						GoPro					
	Deblurring		MFI		Avg		Deblurring		MFI		Avg	
Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
QVI [51]	29.27	0.884	29.98	0.890	29.99	0.890	27.20	0.815	26.62	0.810	–	–
TNTT [21]	29.59	0.889	29.77	0.890	29.75	0.890	26.48	0.808	26.68	0.815	26.65	0.814
DAIN [2]	–	–	32.02	0.901	–	–	–	–	25.54	0.859	–	–
UTI-VFI [56]	28.61	0.879	28.64	0.873	28.64	0.877	25.66	0.808	25.63	0.845	25.64	0.814
EQVI [28]	–	–	29.11	0.893	–	–	–	–	29.68	0.881	–	–
DeMFI [36]	32.90	0.925	32.79	0.926	32.80	0.926	30.82	0.893	30.78	0.900	30.75	0.900
BiT [58]	32.37	0.920	31.88	0.918	31.50	0.915	31.27	0.906	31.20	0.905	28.95	0.885
JNMR [27]	–	–	32.25	0.922	–	–	–	–	31.02	0.900	–	–
IQ-VFI [11]	33.03	0.929	32.90	0.930	32.94	0.928	31.32	0.906	31.26	0.904	31.28	0.904
Ours	33.40	0.932	33.32	0.933	33.35	0.932	32.14	0.912	32.05	0.910	32.08	0.910

4.2. Implementation Details.

We optimize the loss using Adam in the PyTorch framework. The cosine scheduler schedules the learning rate from $1e{-}4$ to $1e{-}6$. Standard data augmentation techniques, such as flipping, rotation, and cropping, are applied to the data with a size of 256×256. We train our model on the training datasets with a batch size of 4 for 900 epochs on 4 NVIDIA 3090ti GPUs.

4.3. Comparison with Previous Methods

As shown in Tab. 1, a total of 12 outperformers are compared, which are fine-tuned on the Adobe and RBI datasets. **Results on the RBI.** Our approach achieves a 1.12dB improvement over the current state-of-the-art. The RBI dataset contains a large number of depth-varying scenes, such as vehicle motion and camera movement, resulting in a high average motion magnitude [58]. DC-BVFI performs well on the RBI dataset, demonstrating that the explicit depth-aware approach is advantageous in challenging scenarios, such as vehicle motion, where many existing methods tend to fail. The proposed DC-BVFI consistently outperforms existing methods, achieving a PSNR of 34.14 on Adobe.

To verify the generalization capability, we conducted cross-dataset experiments using YouTube and GoPro as test sets, as shown in Tab. 2. **Results on the GoPro.** We observed that our method achieved an impressive performance of 32.08 PSNR on the GoPro test set. The GoPro dataset contains a large number of dynamic scenes, resulting in a high average motion magnitude, making it a challenging benchmark for video frame interpolation. Our method performs well on the GoPro dataset, suggesting that DC-BVFI exhibits strong generalization across various motion scenarios. We also observed the final performance of 33.35 PSNR on the YouTube test set.

4.4. Comparison of Visual Results.

We further compare the visualization results in depth-varying motion scenarios, such as car motion and camera motion. As shown in Fig. 4 our method's predictions are

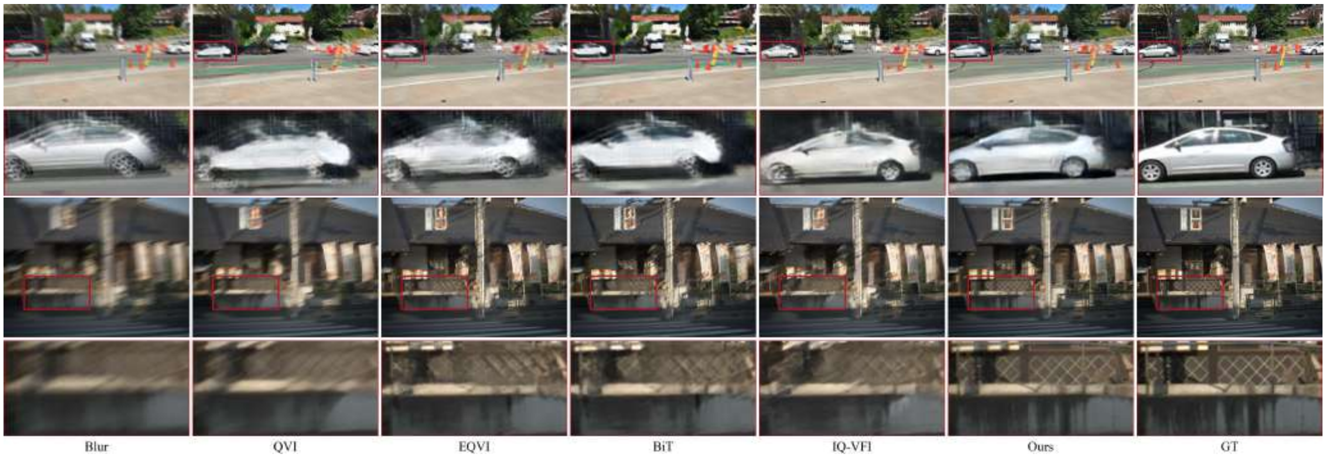

Figure 4. Visual comparison with different methods. The comparison primarily focuses on depth change scenarios, such as car motion and camera movement.

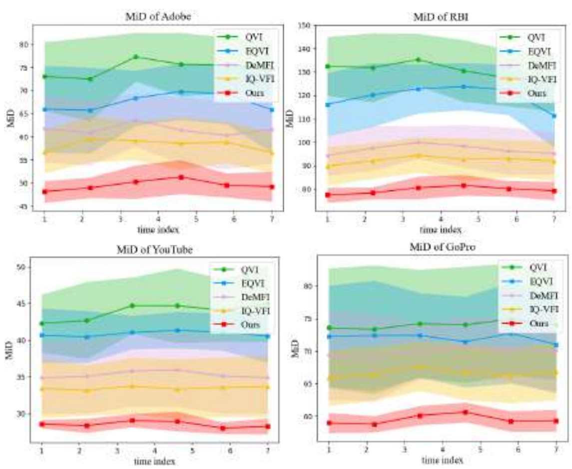

Figure 5. Comparison of Motion-in-Depth.

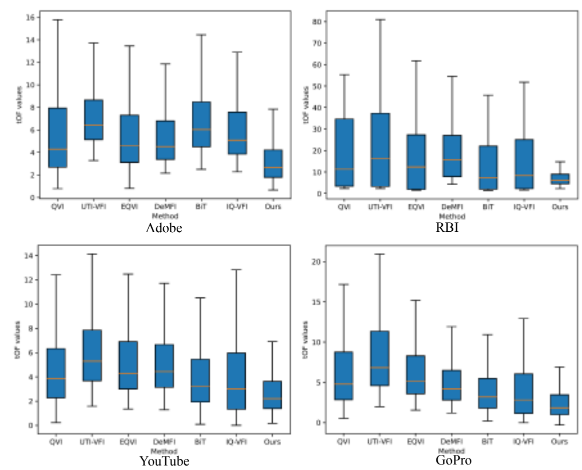

Figure 6. Comparison of temporal consistency performance under tOF metric in Adobe, RBI, YouTube, and GoPro datasets.

closer to the ground truth, with more precise details on both car motion and camera motion. Methods like QVI [51], EQVI [28], and IQ-VFI [11] explore inter-frame motion through estimated acceleration to enhance motion reconstruction. However, they struggle to express changes in object depth during motion, often leading to suboptimal outcomes.

Comparison of Motion-in-Depth. Moreover, to quantitatively measure depth variations during motion. We employ the MiD metric [18] to evaluate the depth variations. Since the video interpolation dataset lacks ground truth depth maps, we use the depth maps estimated from clear video frames [53] as the ground truth labels. Fig. 5 reports the performance of quadratic methods on multi-frame interpolation in Adobe, RBI, YouTube, and GoPro. We can clearly observe that the proposed DC-BVFI method consistently outperforms existing methods in both the mean and variance of MiD. This indicates that our method effectively explores the depth variations cue to enhance video frame interpolation.

Comparison of Temporal Consistency. We employ the tOF metric [7, 16, 41] to evaluate the temporal consistency. Figure 6 reports the quality of motion reconstruction across several existing models on the Adobe, RBI, GoPro, and YouTube test sets. We can clearly observe that the proposed DC-BVFI method consistently outperforms existing methods in both the mean and variance of tOF. The outstanding temporal consistency is due to DC-BVFI's explicit depth modeling of rapid object motion, which greatly improves motion temporal consistency.

4.5. Ablation Studies

We conducted ablation experiments to evaluate the contribution of the 3D architecture and the impact of each submodule in our method, namely the DD and the LSA. Additionally, we investigated the influence of scene flow and optical flow on video frame interpolation. Furthermore, we measured the parameter count and computational complex-

Table 3. Ablation studies of architecture on the RBI dataset. DS stands for stacking video frames along the channel axis, while DD denotes the discriminative deblur strategy. LSA and RB represent Local Self-Attention and Residual Block, respectively.

			UBNet		MPNet		RBI	
Architecture	**BackBone**	**Setting**	DD	DS	LSA	RB	PSNR	MiD(↓)
3D	ViT-Base	**(a)**	✓		✓		31.03	76
		(b)	✓			✓	30.98	83
		(c)		✓	✓		30.66	81
2D	ViT-Base	**(d)**	✓		✓		28.91	137

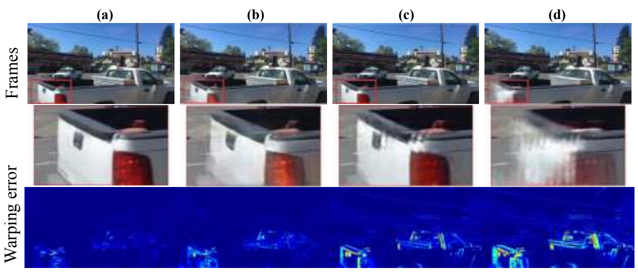

Figure 7. Visualization of the architecture ablation experiments. In the scene with car depth variations, our architecture achieves promising results.

ity of various models.

Architecture. We empirically evaluated the proposed 3D architecture's effectiveness in guiding video frame interpolation in scenes with depth variations. For a fair comparison, both methods use the same backbone but differ in their output prediction heads. PSNR and MiD are presented in Tab. 3. The results of Tab. 3 and Fig. 7 indicate that the 3D architecture significantly outperforms the 2D architecture in scenes with depth variations. The results of setting (b) of Tab. 3 and Fig. 7 highlight the importance of refining scene flow. The results of setting (c) of Tab. 3 and Fig. 7 show that directly stacking misaligned frames hinders intermediate frame restoration, and may not facilitate video deblurring.

Flow. We analyzed the effects of scene flow and optical flow on video frame interpolation performance. For a fair evaluation, in the 2D architecture, we replace the depth estimation in the 3D architecture with a ViT-Base model that has the same parameter size. The results in Tab. 4 and Fig. 8 indicate that scene flow outperforms optical flow in scenarios with depth variations. This is in line with our intuitions, as scene flow integrates depth information on top of optical flow, providing a more comprehensive consideration of useful cues. In other words, when motion doesn't cause depth variations, the scene flow in the depth spatial dimension is deactivated, degrading scene flow into the optical flow. Essentially, scene flow encompasses optical flow. These additional insights contribute to improved video frame interpolation performance in scenes with depth variations.

Table 4. Ablation studies of Flow on the RBI dataset.

			FlowNet		**Depth**		**RBI**	
Architecture	**Flow**	**Setting**	IFNet	RAFT	MeGe	Dust3R	PSNR	MiD (↓)
3D	Scene Flow	**(a)**	✓		✓		31.03	76
		(b)	✓			✓	30.99	82
		(c)		✓	✓		31.00	79
		(d)		✓		✓	30.86	85
Architecture	**Flow**		IFNet	RAFT	ViT-Base 2D		PSNR	MiD (↓)
2D	Optical Flow	**(e)**	✓		✓		28.80	143
		(f)		✓	✓		28.78	147

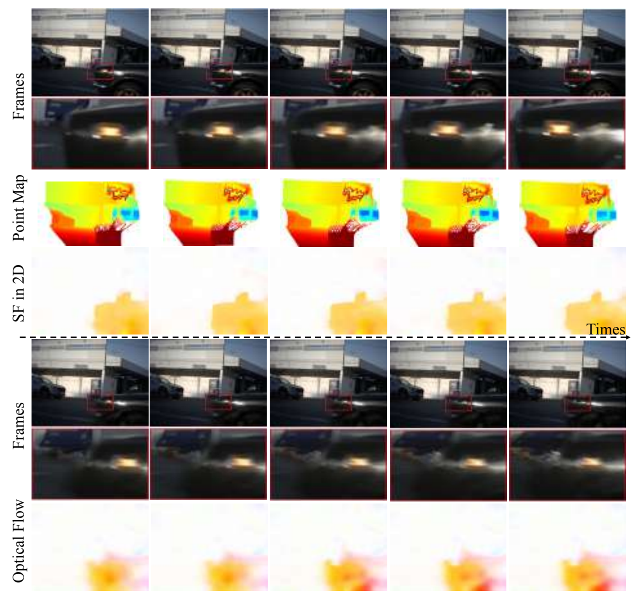

Figure 8. The results of video multi-frame interpolation are presented. The results of the 3D-based architecture (a) are shown in the top rows, and the 2D-based architecture (e) in the bottom rows.

5. Conclusion

This is the first work to focus on solving the depth variations in video frame interpolation. We introduce the differential curves to analyze the issue of depth variations in video frame interpolation. Based on the analysis, this paper builds an explicit depth-aware video interpolation framework to overcome the issue. The UBNet converts video frames into explicit 3D point maps in camera space, and the MPNet estimates the scene flow between these point maps. DC-BVFI has achieved state-of-the-art performance on four benchmark datasets: Adobe, RBI, YouTube, and GoPro.

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